Group #5

Predicting Poverty Susceptibility using U.S. Census Microdata

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***Abstract*—Poverty is a multidimensional public health challenge shaped by demographic, employment, and structural factors. Using microdata from the U.S. Census Bureau's American Community Survey (ACS), we developed machine learning models to identify predictors of poverty risk in California. We constructed a multi-year dataset from the 2018–2024 ACS 1-Year Public Use Microdata Samples (~2.3 million records), excluding 2020 due to data anomalies, and reduced 275 variables to 21 predictors spanning demographics, employment, education, health coverage, disability, and household characteristics. Poverty risk was derived from the Census income-to-poverty ratio (POVPIP) and categorized into four risk levels. Data from 2018–2023 were used for training, with 2024 held out as an out-of-time test set. A multinomial logistic regression baseline achieved a 0.387 average recall across classes and struggled to identify minority poverty-risk groups. Tree-based models modestly improved performance: Random Forest achieved 0.4 average recall, while XGBoost achieved 0.423 average recall for at-risk households. Reframing the task as a binary classification problem (at-risk vs. not at-risk) further improved recall to 0.73 for the Random Forest model and 0.8 for the XGBoost model. Additional exploration of an FT-Transformer architecture and alternative evaluation metrics were conducted during model development. These findings demonstrate how machine learning applied to large census datasets can help identify structural predictors of poverty risk and improve detection of economically vulnerable populations to inform public service programming efforts.**

Keywords—Multinomial Logistic Regression, Random Forest, XGBoost, F-T Transformer, Poverty, Census

I. Introduction

Poverty remains one of the most persistent socioeconomic and public health challenges in the United States. Individuals experiencing poverty face elevated risks of poor physical and mental health, housing instability, reduced access to education, and limited healthcare access. Identifying populations vulnerable to economic hardship is therefore critical for informing public policy, allocating social resources, and designing targeted intervention programs. However, poverty emerges from complex interactions among demographic characteristics, employment conditions, educational attainment, health access, and broader structural socioeconomic factors. Traditional econometric approaches have provided valuable insight into poverty dynamics, but they often rely on smaller datasets and linear assumptions that may fail to capture nonlinear relationships underlying economic vulnerability.

Recent advances in large-scale administrative datasets and machine learning methods provide an opportunity to analyze these dynamics in a more data-driven and scalable way. The U.S. Census Bureau's American Community Survey (ACS) Public Use Microdata Sample contains detailed demographic, employment, housing, and socioeconomic information across the United States. Applying machine learning models to this data allows researchers to identify patterns and predictors of poverty risk that may not be detectable using traditional analytical methods.

In our formulation, the input to the algorithm consists of person-level demographic and socioeconomic indicators derived from ACS microdata, including age, sex, marital status, education level, employment status, hours worked per week, disability status, health insurance coverage, citizenship status, race or ethnicity and language. After exploratory analysis, the original dataset of 275 variables was reduced to a set of 24 predictors designed to capture structural drivers of economic vulnerability while avoiding information leakage. After preprocessing and feature engineering, a final curated set of 21 variables were derived from these 24. Direct income variables were removed because the target variable is derived from income.

These inputs are processed using supervised machine learning models including multinomial logistic regression, Random Forest, and XGBoost to predict an individual's poverty risk category. Poverty status is defined using the Census income to poverty ratio (POVPIP), which compares household income to the federal poverty threshold. We recode this measure into four categories representing different levels of economic vulnerability: Deep Poverty, Poverty, Near Poverty, and Stable. We also evaluate a binary formulation in which individuals are classified as At Risk (POVPIP ≤ 200) or Not At Risk (POVPIP > 200) to address class imbalance and improve detection of vulnerable populations.

Models are trained on ACS data from 2018 through 2023 (excluding 2020) and evaluated on a held-out 2024 test set. This work demonstrates how machine learning applied to large-scale census microdata can help identify structural predictors of poverty risk and support data-informed approaches to identifying populations vulnerable to economic hardship.

II. Related Work

Research on poverty prediction has traditionally relied on econometric and longitudinal analyses to understand the conditions that lead households into or out of poverty. Early work using panel survey data identified key drivers such as employment shocks, household composition changes, and health-related disruptions [7], [6]. A major strength of this research is its ability to examine causal relationships and policy-relevant trends. However, these approaches often rely on relatively small survey samples and linear statistical models, which may struggle to capture complex interactions among socioeconomic variables.

With the increasing availability of large administrative and census datasets, researchers have begun applying machine learning methods to poverty prediction. Studies such as Burger and van der Laan [5] demonstrate that ensemble machine learning models can outperform traditional econometric approaches when predicting poverty transitions. Similarly, Li et al. and Venkatramolla [3], [8] show that algorithms such as random forests and gradient boosting can capture nonlinear relationships between demographic indicators and poverty outcomes using survey data. These studies highlight the strength of machine learning in modeling complex patterns across many predictors, although many are conducted using region-specific datasets, which may limit generalizability. Although our work focuses on California as a target, the overall modeling architecture is primed to scale for assessing other geographic locations in the U.S..

Another line of work focuses on poverty mapping and spatial prediction. Corral et al. [2] show that machine learning can improve poverty estimation by integrating census and geospatial data, particularly in areas where direct measurement is limited. Related work by Hu et al. [1] demonstrates that predictive analytics can improve modeling of social vulnerability and food insecurity. While these approaches provide valuable insights for regional policy planning, their focus on geographic aggregation often limits the ability to identify individual-level predictors of poverty risk.

Machine learning has also been applied to related socioeconomic outcomes such as food insecurity. Okolie et al. [4] compare logistic regression, random forests, and gradient boosting models for predicting food insecurity across U.S. census tracts and find that ensemble methods significantly improve predictive accuracy. These findings support the broader consensus that tree-based models perform particularly well on tabular socioeconomic datasets and support our approach outlined in this project.

More recently, deep learning architectures designed for tabular data have emerged. Gorishniy et al. [11] propose the Feature Tokenizer Transformer (FT-Transformer), which uses attention mechanisms to model complex interactions among structured features. Although these models show promise, they remain computationally intensive and less interpretable than tree-based ensemble methods.

Overall, the literature suggests that machine learning models, particularly ensemble tree-based methods, are effective for predicting socioeconomic outcomes because they capture nonlinear relationships that traditional statistical models may overlook. However, much of the existing research focuses on survey-based poverty transitions or aggregated geographic estimates. Our work extends this literature by applying machine learning models to large-scale individual-level microdata from the American Community Survey to predict poverty susceptibility using structural demographic and employment indicators while intentionally excluding income variables to avoid target leakage.

III. Dataset and Features

Our dataset comes from the American Community Survey (ACS) 1-year Public Use Microdata Sample (PUMS) provided by the U.S. Census Bureau. For this project, we used California person-level PUMS data from 2018, 2019, 2021, 2022, 2023 and 2024, excluding 2020 due to the significant anomalies caused by the COVID-19 pandemic. Across these years, the dataset contains approximately 2.3 million observations and 275 shared variables (including the year). To construct the dataset, we built an ingestion pipeline that downloads the files directly from the Census FTP server. Each yearly dataset is extracted locally and processed so that only the columns shared across all years are retained. The resulting datasets are concatenated into a single master table, observations for individuals younger than 18 are dropped, and a year column is added to keep temporal information. The result is about 2 million records describing demographic characteristics, education, employment, and many other socioeconomic indicators. The modeling problem is framed as a supervised classification task with the variable POVPIP (poverty to income ratio) as the target.

POVPIP measures household income relative to the federal poverty threshold on a percentage scale ranging from 0 to 500. This is a closed range, so there were no outliers to handle in the dataset. To make the prediction task more concrete, we converted POVPIP into 4 classes:

- Deep poverty: 0-50 %
- Poverty: 51-100 %
- Near Poverty: 101-200 %
- Stable: 200+ %

And, into a binary split:

- At Risk: 0-200 %
- Stable: 200+

Extensive exploratory data analysis and preprocessing were conducted to produce the final features and data used in our models. The raw data contains numerous administrative and allocation variables that were not useful for predictive modeling, so we removed them as a result. This includes allocation flags, person weight replicates and identifier fields (trimmed to 115). We also got rid of variables with greater than 60% missing values (trimmed to 94) and variables with extremely high cardinality or > 95% single value dominance (trimmed to 73). We further narrowed down to 24 predictors based on correlation analysis with POVPIP. We also removed all direct income and benefit variables such as wages and retirement income to prevent data leakage, since the target variable itself was derived from income. Instead, we attempted to focus on structural indicators and

demographics. Analysis using 20+ visualizations revealed strong associations between our derived variables and poverty risk.

The 24 predictors at the end of our EDA notebook were further curated to our needs using feature engineering, dimensionality reduction, and preprocessing. Some of these transformations include: aggregating multiple disability indicators into a single disability severity score, collapsing 3 health insurance indicators into a single binary variable (has insurance vs not), one hot encoding race and ethnicity, and collapsing the education tiers representing levels such as high school diploma, some college and bachelor's degree or higher, rather than a continuous numeric measure. These transformations reduced the cardinality of the feature space while preserving meaningful socioeconomic structure, resulting in a more compact and interpretable feature set suitable for downstream modeling.

Missing values were handled using domain aware recoding, for example ENG, WKHP, WRK missing values were replaced with 0 and OCCP missing values were replaced with "NILF", indicating individuals not in the labor force. This approach preserves the semantic meaning of missing values in the dataset rather than replacing them with distribution-based estimates.

After feature engineering, dimensionality reduction was performed using XGBoost feature importance combined with Macro F1 threshold, and supplementary elbow & silhouette analysis. We intentionally retained race, sex, and disability variables to ensure the model captures structural inequalities that may influence poverty risk. Excluding these variables could obscure important social dynamics and reduce the model's ability to identify vulnerable populations. This process identified our final set of 21 features that are used in training all our models.

Our final list of features are: WKL (when last worked), education_tier (engineered education level), MSP (married/spouse present status), WKHP (usual hours worked per week), ESR (employment status), LANX (language other than English spoken at home), MAR (marital status), ENG (ability to speak English), CIT (citizenship status), is_latinx (Hispanic/Latino indicator), MIG (mobility status), has_insurance (health insurance indicator), AGEP (age), SEX, six domain-included race and ethnicity variables: race_white, race_black, race_asian, race_indigenous, race_other, and race_ethnic_aggregate, and disability_score (domain-included disability severity score).

The dataset has extreme class imbalance, with approximately 76% of observations classified as "stable". This imbalance can bias models toward predicting the majority class, so mitigation strategies were required during training. Random under sampling and class weighting were used to address this issue. No normalization was applied for tree-based models, while StandardScaler was applied for our baseline logistic regression.

Given how rich our dataset is, several predictive tasks were possible. Binary poverty classification could predict whether an individual is at risk of poverty or stable and can be used as a screening tool for social services. Multiclass poverty prediction classifies individuals into 4 categories, providing extra granularity for social services with the same use case. Other options included: regression analysis to predict the continuous income to poverty ratio, health insurance coverage prediction, and employment status prediction to address important socioeconomic factors. Given these options, we ultimately focused on the first 2: binary and multiclass poverty risk classification.

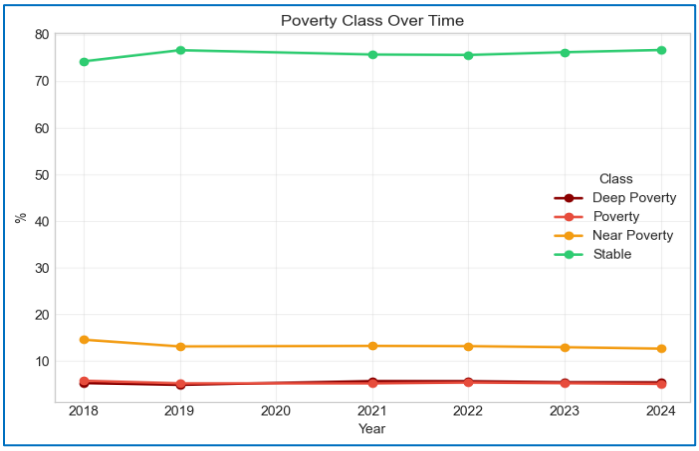

Figure 1: Poverty Classes shown over time 2018-2024 excluding 2020.

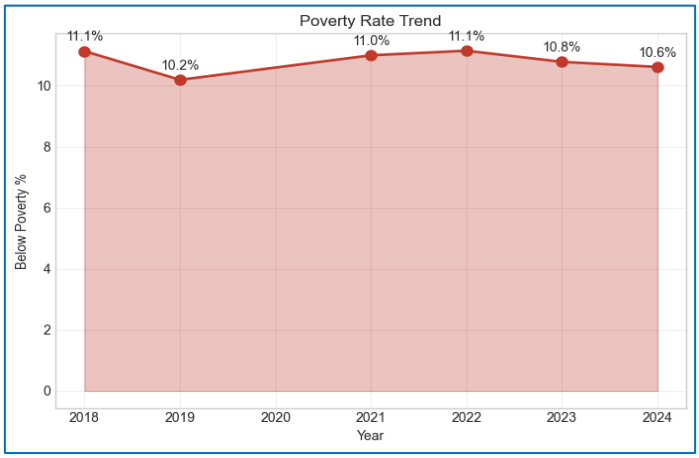

Figure 2: Overall poverty rate across the years 2018-2024 excluding 2020.

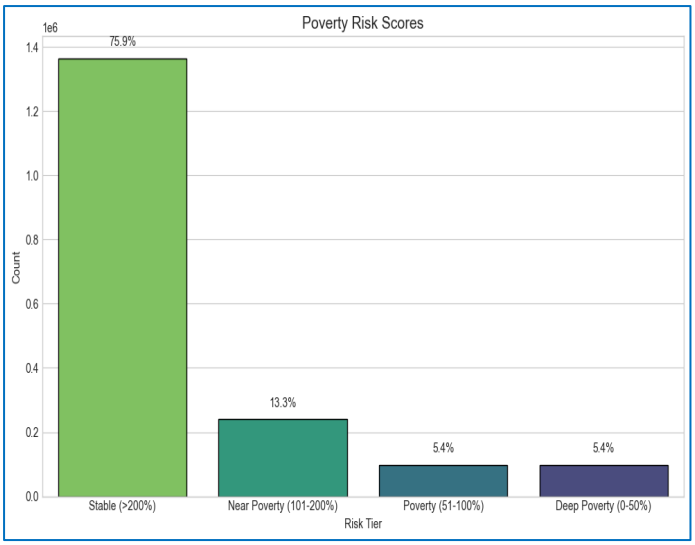

Figure 3: Poverty Classification Distribution.

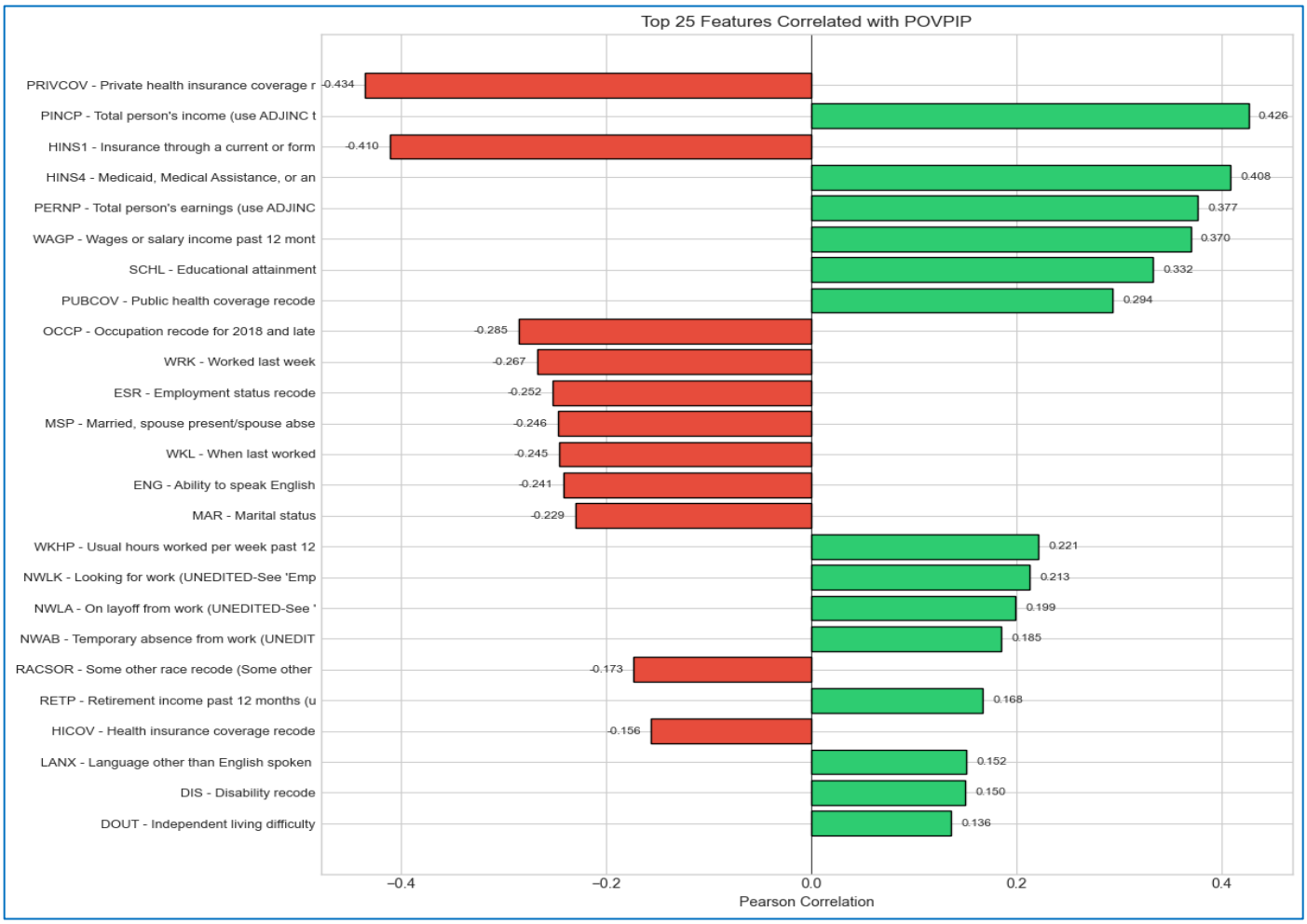

Figure 4: Top 25 features correlated with our poverty target variable POVPIP.

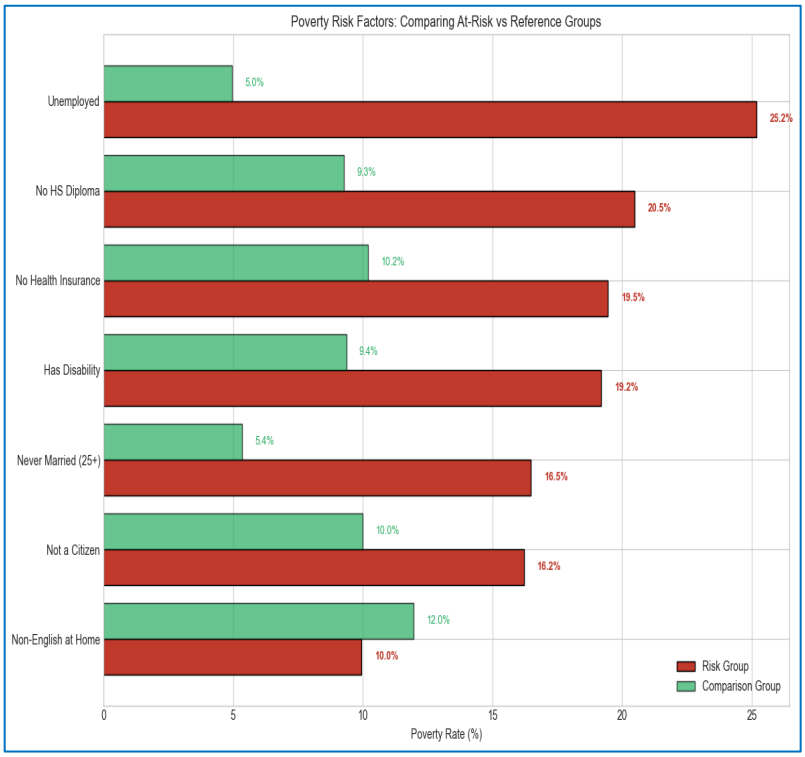

Figure 5: Explored features within poverty versus stable group.

IV. Methods

A. Baseline Model

We trained a multiclass logistic regression model with L2 regularization with C = 0.1 where C controls regularization strength (smaller C = stronger penalty), and the SAGA (Stochastic Average Gradient Descent) solver, which is optimized for large datasets with many features and well-suited to our ~1.47M sample training set. Class imbalance was addressed via the class_weight = 'balanced' setting, which reweights each class inversely proportional to its frequency.

This model was selected for several reasons, it is computationally inexpensive and makes target leakage easy to detect early in the pipeline; it helps surface linear relationships between predictors and poverty-risk class membership before moving to more expressive models; and its learned coefficients are directly interpretable, allowing us to quantify the directional influence of each demographic and socioeconomic predictor. Together, these properties make logistic regression a natural and principled starting point against which Random Forest and XGBoost results can be meaningfully benchmarked.

B. Random Forest (RF)

Random Forest is an ensemble learning algorithm used for supervised classification and regression tasks. It constructs multiple decision trees using bootstrap samples of the training data and random subsets of features at each split, which introduces randomness that reduces variance and helps

prevent overfitting [12]. For classification problems, each tree produces a prediction, and the final output is determined by majority vote across all trees [12]:

$$\hat{y} = \text{majority vote}\{T_1(x), T_2(x), \ldots, T_K(x)\}$$

Where $T_k(x)$ represents the prediction from the k-th decision tree and k is the total number of trees in the ensemble. Because each tree is trained on different subsets of observations and features, Random Forest models can capture nonlinear relationships between predictors while maintaining strong generalization performance.

For this implementation, the model was trained using the preprocessed ACS dataset containing demographic, employment, education, and household indicators. Class imbalance was addressed using the *balanced* and *balanced_subsample* weighting strategies, which adjust class weights inversely proportional to class frequencies. The model was trained on ACS data from 2018–2023 and evaluated on the held-out 2024 test set using accuracy, precision, recall, and F1-score, with particular emphasis on recall for poverty-risk classes.

C. XGBoost

Extreme Gradient Boosting (XGBoost) is a very efficient gradient boosting algorithm designed for supervised learning on large tabular datasets. It builds an ensemble of decision trees sequentially, where each new tree attempts to correct the errors made by previous trees. This boosting process allows the model to capture complex nonlinear relationships between predictors and the target variable [9]. They work by minimizing an objective function that combines prediction loss with a regularization penalty to control model complexity [10]. The general objective function used in XGBoost is:

$$\text{obj}(\theta) = \sum_{i}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \omega(f_k)$$

Where $l(y_i, \hat{y}_i)$ represents the loss function measuring the difference between the predicted value $\hat{y}_i$ and the true label y_i, and f_k represents the k-th decision tree in the ensemble. The regularization term $w(f_k)$ penalizes overly complex trees to reduce overfitting and improve generalization [10].

The regularization component is defined as:

$$\omega(f) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2$$

Where T is the number of leaves in the tree, w_j represents leaf weights, and $\boldsymbol{\gamma}$ and λ control the complexity penalty applied to the model. This is one of the key reasons XGBoost performs well on high-dimensional datasets.

XGBoost is great for this project because the census data contains a large number of features including categorical, binary, and continuous socioeconomic factors. Poverty risk is influenced by complex interactions between these predictors. Tree based models are able to automatically model these nonlinear relationships without requiring extensive manual feature transformations. XGBoost is also computationally efficient and scalable because it includes several engineering optimizations such as parallelized tree construction or efficient handling of sparse data. These features make it effective for large datasets like the census.

D. The Feature Tokenizer Transformer

The Feature Tokenizer + Transformer (FT-Transformer) is a deep learning architecture proposed by Gorishniy et al. [11] specifically designed for tabular data, where standard transformers historically underperformed tree-based methods The model first maps every input feature, both categorical and continuous, into a shared embedding space of dimension d via a tokenization step [11]:

$$T_j^{(num)} = b_j^{(num)} + x_j^{(num)} \cdot W_j^{(num)} \quad \in \mathbb{R}^d,$$
$$T_j^{(cat)} = b_j^{(cat)} + e_j^T W_j^{(cat)} \quad \in \mathbb{R}^d,$$
$$T = \texttt{stack}\left[T_1^{(num)}, \ldots, T_{k^{(num)}}^{(num)}, T_1^{(cat)}, \ldots, T_{k^{(cat)}}^{(cat)}\right] \in \mathbb{R}^{k \times d}$$

Continuous features $x_j^{(num)}$ as shown in the formula in $T_j^{(num)}$ where $W_j^{(num)}$ is a learned weight vector and b_j is a learned bias. Categorical features $x_j^{(cat)}$ is embedded via a lookup table and described through the formula $T_j^{(cat)}$ where e_j is a one-hot encoded vector and b_j is a learned bias projecting each scalar feature value into the shared embedding space via element-wise multiplication. This gives every feature, regardless of type, a dense vector representation in the same R space. A special [CLS] token is prepended to the resulting token sequence and passed through $L = 3$ Transformer encoder layers, each applying multi-head self-attention with n = 4 heads to learn pairwise feature interactions, after which all token representations are flattened and passed to a two-layer Multi-Layer Perceptron (MLP) classification head. The MLP applies the ReLU activation function, defined as ReLU(z) = max (0, z), which introduces non-linearity by zeroing out negative values and allowing the network to learn complex decision boundaries [11]:

$$\hat{y} = \texttt{Linear}(\texttt{ReLU}(\texttt{LayerNorm}(T_L^{[\texttt{CLS}]})))$$

We chose FT-Transformer because its self-attention mechanism explicitly models pairwise interactions between all features at every layer, a particularly valuable property for poverty prediction, where the joint effect of features like employment status, education, and citizenship is more predictive than any individual feature alone. Unlike tree-based models that partition the feature space through splits, the transformer learns soft, continuous feature dependencies, making it well-suited to the mixed categorical/continuous structure of ACS microdata. Recent benchmarks [1] confirm FT-Transformer matches or exceeds gradient-boosted tree performance on tabular datasets of this scale and complexity, justifying its inclusion alongside our tree-based baselines.

V. EXPERIMENTS/RESULTS/DISCUSSION

ACTUAL VALUES	POSITIVE	NEGATIVE
POSITIVE	TP	FN
NEGATIVE	FP	TN

$$Precision = \frac{TP}{TP + FP} \qquad Recall = \frac{TP}{TP + FN}$$

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN}$$

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

A. Baseline Model

For the Baseline model, we evaluated on the 4-multiclass target variable Hyperparameters were set as follows: L2 regularization strength C=0.1 (selected to apply moderate penalization and reduce overfitting on the high-dimensional feature matrix), class_weight = 'balanced' to reweight classes inversely proportional to their frequency, and the SAGA solver with max_iter = 10000 to ensure convergence on the ~1.49M training samples. No cross-validation was performed given the temporal train/test split design, where 2018–2023 ACS data was used for training and 2024 data served as the held-out test set. On the 2024 test set, the model achieved an overall accuracy of 60%, a weighted F1 of 0.65, and a macro-averaged ROC-AUC of 0.76 (one-vs-rest). The per-class performance was highly uneven: the Stable class achieved strong precision (0.91) but moderate recall (0.67) and F1 (0.77), while minority poverty-risk classes performed considerably worse where Near Poverty (F1 = 0.25), Poverty (F1 = 0.18), and Deep Poverty (F1 = 0.30). This shows that the model frequently misclassifies Near Poverty and Poverty instances as Stable, reflecting the model's tendency to collapse predictions toward the majority class despite balanced class weighting. The coefficient plot reveals that hours worked per week (WKHP), education tier, and employment status (ESR) carry the largest influence across all four classes. The baseline's results for the minority-class performance, Deep Poverty is expected given logistic regression's linear decision boundary, which cannot capture the complex nonlinear interactions between socioeconomic predictors that likely characterize transitions between poverty-risk categories. These limitations directly motivate the adoption of Random Forest and XGBoost in subsequent sections.

B. Random Forest

The Random Forest model was configured with 200 trees, min_samples_split = 10, and min_samples_leaf = 5, with class weighting used to address imbalance across poverty categories. The number of trees was selected to stabilize ensemble predictions while maintaining reasonable computational cost given the large training dataset. Increasing the number of trees generally reduces variance but yields diminishing returns beyond several hundred trees. The parameters min_samples_split and min_samples_leaf were introduced as regularization controls to prevent overly specific splits that could capture noise in the data. Because the ACS microdata contains millions of observations and heterogeneous demographic characteristics, unrestricted tree growth can produce highly fragmented decision boundaries that do not generalize well. Requiring a minimum of ten samples to split a node and at least five samples per leaf encourages broader partitions of the feature space and improves model stability. Class imbalance across poverty categories was addressed by evaluating both class_weight = "balanced" and class_weight = "balanced_subsample". The balanced_subsample option recalculates class weights within each bootstrap sample used to train individual trees, allowing the model to adapt to variations in class distribution and improve learning for underrepresented poverty classes.

To explore effective model configurations while controlling computational cost, we conducted a RandomizedSearchCV hyperparameter search on a stratified subset of the training data. Approximately 35% of the training set was used for the search procedure to reduce runtime while preserving class proportions. The final configuration reflects a balance between predictive performance, model stability, and computational efficiency observed during this tuning process.

In the four-class poverty prediction task (Stable, Near Poverty, Poverty, Deep Poverty), the Random Forest model achieved an average poverty-class recall of 0.40 and a macro F1 score of 0.40 on the 2024 holdout dataset (n = 308,479). Compared to the multinomial logistic regression baseline (0.39 average poverty recall, 0.38 macro F1), Random Forest provided modest improvements in both recall and overall predictive balance. Class-level performance remained uneven, with Near Poverty recall of 0.41, Poverty recall of 0.28, and Deep Poverty recall of 0.51. While the logistic regression baseline achieved slightly higher recall for Deep Poverty, Random Forest produced more balanced performance across the remaining poverty classes, reflecting the advantage of ensemble tree-based models in capturing nonlinear relationships among demographic, employment, and household characteristics.

Because distinguishing between adjacent poverty categories proved difficult across models, we also evaluated a binary formulation separating Stable households from At-Risk households (Near Poverty, Poverty, and Deep Poverty combined). Under this framing, Random Forest achieved 0.73 recall for the At-Risk class, correctly identifying 52,594 of 71,898 at-risk individuals in the 2024 holdout set while misclassifying 19,304 individuals as stable, and achieved a macro F1 score of 0.69. This binary formulation substantially improved performance relative to the multiclass task, suggesting that ACS socioeconomic indicators are more informative for identifying broad vulnerability to poverty than for distinguishing between closely related poverty thresholds

C. XGBoost

For the XGBoost model, we evaluated both a binary and a 4-class model. Model validation used stratified 5-fold cross validation with random under sampling to address class imbalance. The main model used hyperparameters: max depth = 6, learning rate = 0.1, number of trees = 500, subsample = 0.8, and colsample_bytree = 0.8, with early stopping rounds = 30. These parameters were chosen to balance model complexity and stability. Deeper trees increase the risk of overfitting, while a moderate learning rate allows the model to gradually correct prediction errors. A small grid search over several hyperparameters produced minimal variations in results, so this configuration was kept.

Model performance was evaluated using mainly at-risk recall and F1 score because the goal of the project is to identify vulnerable households. The equations for these metrics are

provided in the figure above. For our use case, missing households that are actually at risk (false negatives) is more harmful than incorrectly flagging stable households.

Cross validation results were stable across folds. On the test set, the best binary model achieved approximately 0.69 F1 and 0.77 recall for the at-risk class. We also performed threshold tuning using the precision-recall curve, selecting a threshold of approximately 0.462 to maximize recall. This model produced our best binary at risk recall of 0.8, successfully identifying 4 in 5 at risk households, though there was certainly the tradeoff of more false positives.

The 4-class model produced lower performance, achieving about 0.4 macro F1 and 0.43 average poverty recall on the test set. Most errors occurred between adjacent poverty categories, e.g. near poverty vs poverty, suggesting that socioeconomic indicators usually overlap between these groups. 3D PCA analysis shows that the 3 lower categories are found in the same regions of feature space, making them quite difficult to separate, and we were working against extremely heavy class imbalance.

Feature importance analysis showed that education tier, marital status, and hours worked per week were among the strongest predictors of poverty risk. Overall, XGBoost performed well because it can capture nonlinear interactions between socioeconomic factors. Confusion matrices comparing the binary Random Forest and XGBoost are shown in Figure 6/Figure 7.

D. FT-Transformer

The FT-Transformer was evaluated on a 4-class poverty severity target (Stable, Near Poverty, Poverty, Deep Poverty) using a temporal holdout strategy on the held-out 2024 survey year, yielding a test set of 308,479 adults

No formal validation fold was used during training; model convergence was monitored via average training loss per epoch. The final model was trained with the following hyperparameters: embedding dimension d=64, AdamW optimizer with learning rate $\eta=1\times10^{-3}$ and weight decay $\lambda=1\times10^{-4}$, mini-batch size of 256, 10 training epochs, cosine annealing scheduler with T_{max} = 10, and gradient clipping at $\|g\|_2 \leq 1.0$

At the default argmax threshold Class 2 (Poverty) achieved a recall of 0.081 with precision of 0.260 and F1 of 0.115, indicating the model correctly identified only 8.1% of true Poverty cases in the 2024 holdout, the weakest of any class.

To investigate whether this was recoverable without retraining, we swept the Class 2 probability threshold from 0.02 to 0.29, predicting Poverty whenever p-hat^2 > t regardless of argmax. The best threshold under a minimum precision constraint of 0.10 was t = 0.08, which improved Poverty recall substantially to 0.832 while raising precision to 0.102 (F1 = 0.181).

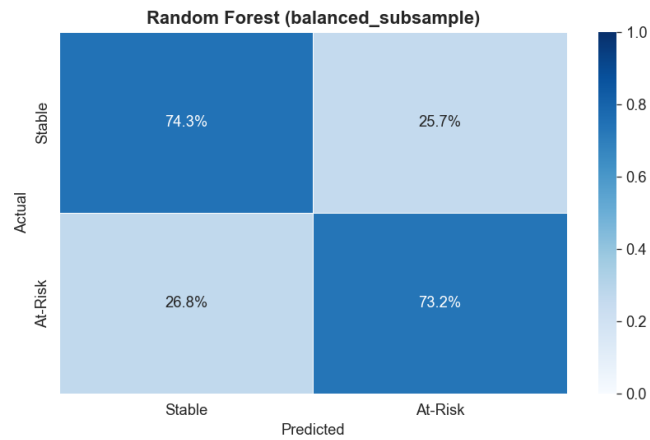

Figure 6: Confusion matrix for the binary Random Forest.

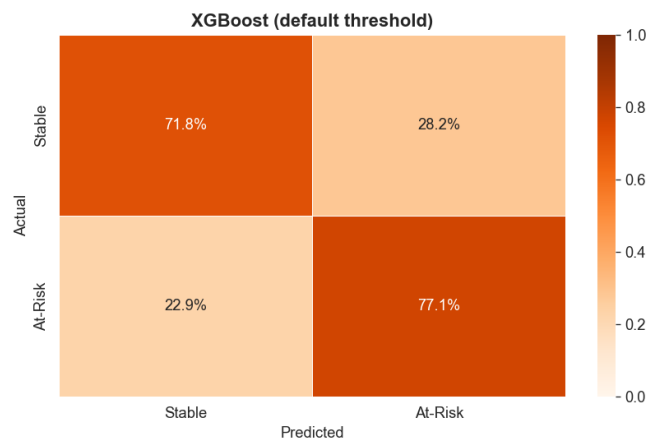

Figure 7: Confusion matrix for the binary XGboost model.

Figure 8: Recall across all models.

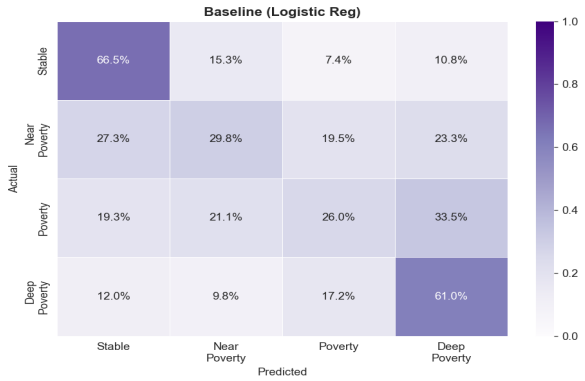

Figure 9: Confusion matrix for the multiclass baseline multinomial logistic regression.

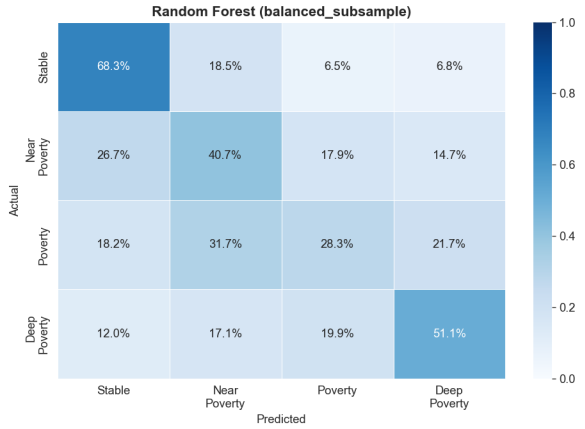

Figure 10: Normalized confusion matrix for the multiclass Random Forest.

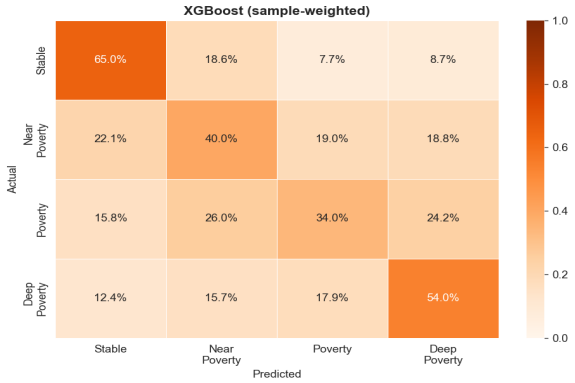

Figure 11: Confusion matrix for the multiclass XGboost.

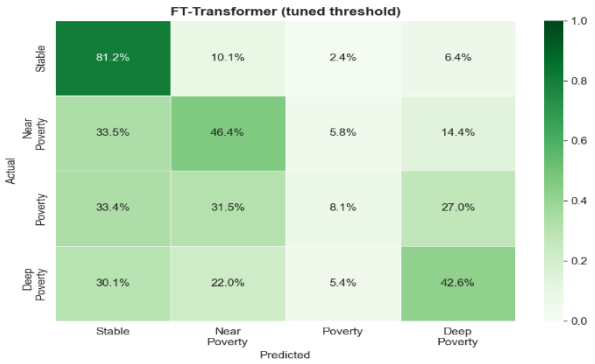

Figure 12: Normalized confusion matrix for the multiclass FT-Transformer.

VI. Conclusion

This study demonstrates how machine learning methods can be applied to large-scale census microdata to identify structural predictors of poverty risk. Using person-level data from the American Community Survey spanning 2018–2024, we constructed a modeling pipeline that reduced hundreds of raw variables to a compact set of interpretable demographic and socioeconomic predictors. Across multiple model architectures, tree-based ensemble methods consistently outperformed the logistic regression baseline, highlighting the importance of modeling nonlinear relationships and interactions among structural factors such as education, employment, and household composition.

Our findings show that Random Forest and XGBoost provide meaningful improvements in detecting economically vulnerable populations, particularly when the prediction task is reframed as a binary classification problem focused on identifying individuals at risk of poverty. In this setting, the XGBoost model achieved the strongest performance, identifying approximately 80% of at-risk households under a high-recall configuration. The FT-Transformer, while theoretically expressive, did not overcome the challenge of deep learning on structured tabular data with class imbalance on a 4-class categorical target variable. These results suggest that machine learning models trained on census microdata can serve as useful screening tools for identifying populations that may benefit from targeted social programs or policy interventions.

At the same time, several limitations should be acknowledged. Poverty categories defined by income thresholds often overlap in the feature space, making it difficult for models to distinguish between adjacent classes such as Poverty and Near Poverty. This brings into question how poverty is defined within the census in an evolving economic environment, which we think should be further explored. This raises the question of whether the current definition for poverty adequately captures the diversity of experiences in the lower socioeconomic classes. Additionally, the models rely on cross-sectional survey data rather than longitudinal observations, limiting the ability to capture causal dynamics or predict transitions into and out of poverty over time. Future work could incorporate temporal modeling, geographic features, or additional administrative datasets to improve predictive performance and better capture structural determinants of economic vulnerability.

From a societal perspective, the ability to identify economically vulnerable populations using publicly available census data has significant implications for policy design and resource allocation. A model that correctly flags 80% at-risk households using structural covariates, without relying on direct income disclosure could support early intervention programs, targeted benefit eligibility screening, and geographic prioritization of social services. The prominence of education tier as a key predictor reinforces established empirical findings on the poverty-protection effects of educational attainment, suggesting that upstream investments in educational access may be among the most effective poverty-prevention strategies. Similarly, the predictive weight of hours worked per week, and

employment status highlights the structural role of labor market conditions.

Overall, this work highlights the value of combining large public datasets with modern machine learning techniques to better understand socioeconomic risk. By focusing on structural predictors rather than direct income variables, the models provide insight into the underlying conditions associated with poverty susceptibility and demonstrate how data-driven approaches can complement traditional economic analyses in identifying vulnerable populations.

VII. Contributions

Ingrid: Contributed to the GitHub setup including READme file, and conda environment setup. Created script for data ingestion steps. Added to exploratory analysis enhancing initial findings. Implemented multinomial logistic regression baseline model using 4 categorical variables. Conducted research and implementation on the FeatureTokenizer Transformer. Completed several rounds of model evaluation and performance.

Marianne: Contributed to exploratory data analysis, modeling, and project infrastructure. Contributed to EDA and preprocessing, including analysis of visuals, feature distributions/engineering, and correlation structures to aid in identifying predictors. Implemented the XGBoost model pipeline, including several variations of the model, and model evaluation. Played a role in organizing and structuring GitHub repository and contributed to writing sections of the reports.

Vanessa: Contributed to the modeling preprocessing pipelines and notebook collation, helped with initial EDA creation and exploratory targets. Implemented the Random Forest models with corresponding analyses, model evaluation and tuning. Significant contribution to all submission report structures, content formatting, and editing. Played a role in organizing and structuring GitHub repository and contributed to writing sections of the reports.

VIII. Reflections

If given the opportunity to redo this project, there are several things we would approach differently, though the dataset and core problem would remain the same. We are all genuinely passionate about poverty as a public health issue, and the American Community Survey remains one of the richest publicly available sources for studying economic vulnerability at scale.

That said, given the breadth of the dataset at 275 columns across multiple years, we would invest significantly more time upfront in data processing and feature engineering. With more time, we would have explored further feature aggregation strategies, combining related columns into composite indicators to reduce dimensionality while preserving signals. We also wish we had more deeply interrogated features with potential for bias, particularly race, nationality, and ethnicity. These variables have real predictive signals but understanding how and why they influence model output is important for responsible interpretation and any downstream policy application, that analysis deserved more dedicated attention than we were able to give it.

On the modeling side, we would have established separate baseline models for each target variable encoding strategy. Our target variable evolved over the course of the project, starting as a four-class poverty risk score (Deep Poverty, Poverty, Near Poverty, Stable) before being binarized to a simpler at-risk vs. stable classification. Comparing later models against a baseline built on a different encoding made fair evaluation difficult; in hindsight, we would have set baselines for both formulations from the start.

From a technical infrastructure standpoint, working within the DataHub environment introduced low memory constraints that we didn't fully anticipate. Performance limitations pushed us toward migrating to Parquet file formats toward the end of the project to successfully convert the data to tensor format, which added overhead. A more deliberate environment setup, including better planning around tensor conversion and data storage formats, would have saved considerable time. We also would have engaged earlier with domain side guidance, as those conversations consistently sharpened our thinking and pushed the analysis in more meaningful directions.

Finally, we briefly considered supplementing the ACS data with housing data, which could have added an important structural dimension to our poverty model. That remains an avenue worth pursuing in any future iteration of this work

IX. Reply to review

We appreciate the thoughtful feedback response provided by our reviewing groups. Below, we summarize how we addressed or considered each point in the final version of the project.

One of our reviewers suggested including a single visualization comparing the performance of all models. In response, we added comparative result summaries and confusion matrices across our models in a final summary notebook. Another comment raised concerns about the imputation strategy used for missing values. We revisited this section and clarified our preprocessing approach, ensuring that the handling of missing values was more clearly documented in the final report. Finally, the first reviewer suggested expanding the ethical considerations of poverty prediction, particularly regarding the consequences of false positives and false negatives. We incorporated additional discussions in the report regarding ethical implications. In particular, we emphasize that false negatives (failing to identify a family at risk) may be more harmful than a false positive (services going to relatively more stable households) in the context of social services, which is why our models prioritize recall for vulnerable families.

Our second group of reviewers noted that our model may have limited generalizability because it was trained on California data and suggested expanding the analysis to include other states. While we agree that evaluating the model across multiple states would provide a stronger test of generalizability, we intentionally limited the dataset to California for the scope of this project due to limits in compute power, and also to control for differences amongst states. The ACS PUMS dataset is nationally available, and our data ingestion pipeline was designed to easily extend to other states in a step for future work. Another suggestion involved adding gender back into the feature set, we recognized the value of this and added this step in our preprocessing, so sex feature was used in all of our models. Finally, they recommended exploring interaction features and intersectional indicators that

may capture compounding risks. While our current models focus primarily on core demographic and employment indicators, we agree that engineered interaction variables may represent an important direction for future work.

Overall, the reviewers' feedback helped us look at our project from a different perspective. They helped us strengthen the clarity of our methodology, expand our discussion of ethical implications, and consider additional engineering approaches that could improve future iterations of the models.

ACKNOWLEDGMENT

The authors thank LaKeta Kemp for her domain expertise and guidance. We also express our gratitude to the instructional team of DSC 288R Capstone Project.

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Key libraries used in this project include NumPy [13], Pandas [14], and SciPy [15] for data preprocessing; Scikit-learn [16] and PyTorch [17] for machine learning and model development; and Matplotlib [18], Seaborn [19], and JoyPy [8] for visualization.

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